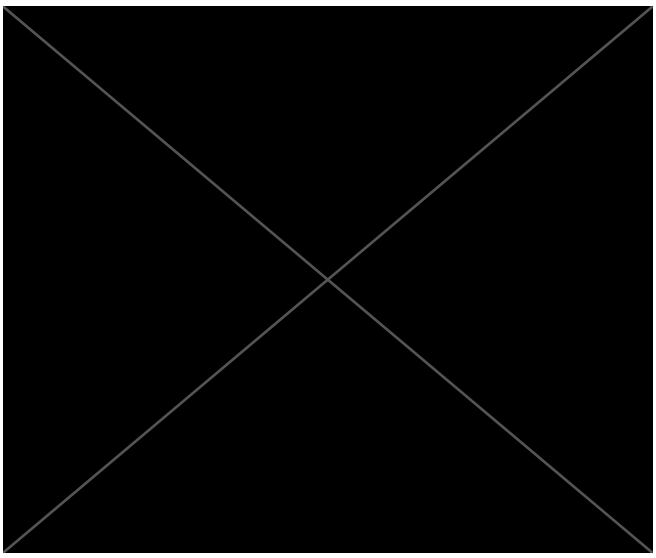
Mechanistic Interpretability of Tissue-Specific Gene Expression Prediction in Transformers

Project Report



June 30, 2026

Abstract

We investigate whether Transformer models trained to predict tissue-specific gene expression from DNA sequence develop internal routing mechanisms driven by a tissue-context token. To separate predictive performance from recovery of the intended conditional rule, we first constructed a synthetic task in which the motif `GATTACAA` increased expression only in the `HEAD` tissue. Character-level tokenization failed as sequence length increased, while overlapping 6-mers improved local motif recognition. A hierarchical local-to-global Transformer further enabled reliable aggregation over 2200-nucleotide inputs: the best hierarchical model achieved normalized test MSE 0.240 ± 0.016 and recovered $95.9\% \pm 1.3\%$ of the known interaction, compared with 0.595 ± 0.054 MSE and $7.8\% \pm 9.0\%$ recovery for the flat 6-mer Transformer. On real Drosophila data, tissue-token interventions and DNA perturbations showed that the models relied on both tissue identity and gene-specific sequence information. Causal activation patching indicated that the hierarchical model used distributed, predominantly MLP-based routing, whereas the MLM-pretrained flat Transformer showed a more localized head-level effect for a strongly separated tissue pair. Attention rollout and 6-mer occlusion recovered the known synthetic motif but did not identify a stable real-data motif. These results demonstrate that low prediction error alone is insufficient for mechanistic validation and that synthetic controls are essential for evaluating tissue-conditioned genomic models.

1 Introduction

A Mixture of Experts (MoE) layer routes specific tokens to specialized sub- networks to save computation and increase efficiency by specializing MoE heads. It is also in biology: every cell in the human body has the exact same DNA (the input sequence is universal), but a liver cell routes that sequence through "liver logic" to produce liver proteins, while a neuron routes it through "brain logic". The model must learn to conditionally activate entirely different sub-networks of its weights based on a single "context" token. We want to train a transformer model (without MoE layer) on a dataset which should naturally force the model to learn such specialized rerouting. The genome of an organism can be, in a simple picture, understood as a large single line string in which we have buried words (genes) which have a meaning. Some words (genes) are used in one part of the organism (like stomach which needs genes producing stomach acid) or brain which is the only place where a human organism has a large concentration of neuron cells.

2 Related Work

Sequence-based prediction of gene expression has been studied extensively. Early deep learning models such as DeepSEA (Zhou and Troyanskaya, 2015) and Basset (Kelley et al., 2016) showed that neural networks can learn regulatory sequence features directly from DNA. Later models, including Basenji (Kelley et al., 2018), ExPecto (Zhou et al., 2018), and Xpresso (Agarwal and Shendure, 2020), moved towards larger-scale and more direct prediction of gene activity. More recently, transformer-based models such as Enformer (Avsec et al., 2021) and Transformer DeepLncLoc (Pipoli et al., 2022) demonstrated that attention-based architectures can be useful for predicting gene expression from sequence.

Transformers have also been used to analyze DNA sequences and gene expression data. Large DNA models such as Nucleotide Transformer (Dalla-Torre et al., 2025) and DNABERT-2 (Zhou et al., 2024) show that transformers can learn useful features from genome sequences. Other works study how to interpret genomic transformers, for example by analyzing internal representations and attention heads (Clauwaert et al., 2021; Consens et al., 2025).

Our work is additionally motivated by Mixture-of-Experts models, where learned gating mechanisms route tokens to specialized subnetworks (Shazeer et al., 2017; Fedus et al., 2022). In contrast, our model does not include an explicit MoE layer. Instead, we ask whether a dense transformer trained on tissue-conditioned genomic inputs develops an emergent routing-like mechanism. While prior work has studied gene expression prediction and interpretability of genomic transformers, it remains unclear whether tissue-conditioned biological tasks induce internal specialization similar to Mixture-of-Experts routing.

3 Data and Experimental Setup

3.1 Real data

The foundational dataset maps raw Drosophila melanogaster (w1118) genomic sequences to continuous expression values across seven distinct tissues (Abdomen, Digestive, Genitalia, Gonads, Head, Reproductive, Thorax). Raw RNA- seq read counts were sourced from the NIH GEO database (accession GSE99574). As illustrated in figure 1, the data pipeline consists of two parallel branches—count normalization and genomic sanitization —that merge to form the final string-count tensor. Raw RNA-seq counts suffer from severe sampling bias due to varying sequencing depths and inherent variance scaling with the mean. To reframe these counts as stable continuous targets for a regression head, we applied a two-step statistical correction using Median of Ratios which scales the data against a geometric pseudoreference preventing highly expressed genes from artificially silencing less active ones.The second step is the Variance Stabilizing Transformation (VST): A non-linear transformation that decouples variance from the mean preventing massive count outliers from dominating the Mean Squared Error (MSE) loss. To construct the input space (X), the normalized expression targets were mapped to their physical DNA sequences using the Drosophila reference genome and annotations (FB2017 03 release 6.16). From each gene we extracted 1000 nucleotides of its promoter region and entire 5'UTR regulatory region which are characteristic parts of the gene sequence. The concatenated length was capped at 2200 characters, discarding all longer sequences.

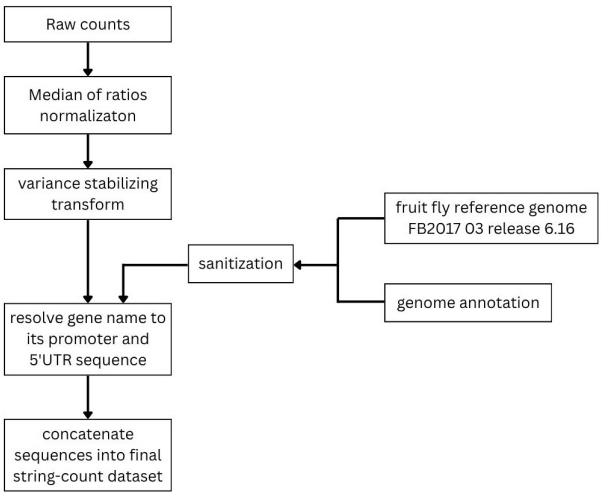


Figure 1: Bioinformatic data processing pipeline. The final dataset consists of string of DNA sequences paired with counts and source tissue.

The production dataset constitutes a one-to-many mapping of sequence-to- tissue expression. It contains 79,905 total samples, perfectly balanced across the 7 tissue contexts (14.3% per tissue). For all downstream tasks, the data was partitioned into strict 80/10/10 splits for training, validation, and testing, respectively. Dataset statistics are presented in figure 9.

3.2 Synthetic data

To decouple model development from the complex standardization of raw Drosophila genome annotations, we constructed a synthetic dataset. This provided a deterministic environment to validate tensor shapes, verify gradient flow, and benchmark the architecture before introducing noisy biological data. The input space (X) approximated real genomic dimensions by concatenating a fixed 1,000-bp promoter with a variable 50–250 bp 5' UTR. To mimic the natural biological distribution of fly promoters, background sequences were generated using an AT-rich grammar (35% A, 35% T, 15% C, 15% G). To verify the network actively utilized the conditional [TISSUE] token, target expression values (y) were governed by a strict causal rule:

1. Basal State: Default expression was drawn from a low-level continuous Gamma distribution.

2. Activated State: A specific 8-mer motif (GATTACAA) functioned as a causal trigger. If this motif was present AND the token was [TISSUE=HEAD], the target was mathematically forced into a high-expression state.

To prevent the model from exploiting heuristic shortcuts (e.g., merely associating the motif with high expression), the GATTACAA sequence was injected into 80% of all samples, regardless of their assigned tissue. This negative control guaranteed the network could only minimize loss by computing the joint interaction between the sequence motif and the specific tissue context.

3.3 Evaluation Protocol

Synthetic performance was evaluated in four groups defined by tissue identity and motif presence: `HEAD` with or without the motif, and all other tissues with or without the motif. In addition to MSE, models were compared with constant and tissue-only baselines. Recovery of the known synthetic rule was measured by

$$I_{\text{pred}} = \left(\bar{\hat{y}}_{\text{HEAD,motif}} - \bar{\hat{y}}_{\text{HEAD,no motif}}\right) - \left(\bar{\hat{y}}_{\text{other,motif}} - \bar{\hat{y}}_{\text{other,no motif}}\right),$$

with I_{target} defined analogously from the targets. Low MSE with near-zero I_{pred} was treated as shortcut learning. A balanced-sampling control did not improve interaction recovery.

All final comparisons used fixed held-out partitions shared across model variants. Real-data splits were grouped by gene, and sequences longer than 2200 nt were removed at the gene level. The real test set contained 1,142 genes and 7,994 records; the synthetic 2200-nt test set contained 300 sequences. Normalization and baselines were fitted on training data only, checkpoint selection used validation MSE, and principal comparisons used seeds 42–45. Unless stated otherwise, results are reported using normalized MSE. Early stopping and gradient clipping were applied during training.

Interaction recovery is reported only for synthetic data, where the target interaction is known. On real data, the corresponding target difference-in-differences was approximately 0.026 in the raw scale, making recovery ratios unstable; the statistic is therefore used only descriptively.

4 Models

4.1 Diagnostic-Guided Architecture Development

The synthetic diagnostic exposed two successive bottlenecks. Character-level tokenization failed to preserve the tissue–motif interaction as sequence length increased. Overlapping 6-mers, inspired by DNABERT (Ji et al., 2021), solved the local representation problem but retained an almost unchanged sequence length, motivating hierarchical local-to-global compression (Figure 2).

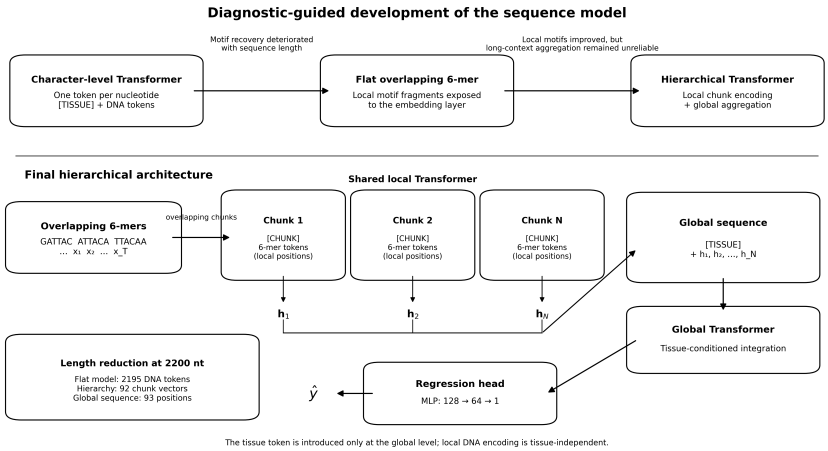


Figure 2: Diagnostic-guided development of the model architecture. The upper panel shows the sequence of architectural changes and the failure motivating each change; the lower panel presents the final hierarchical model.

The final model compresses overlapping 6-mer chunks with a shared local Transformer and combines the ordered chunk representations with the tissue token in a global Transformer. The contextualized tissue representation is passed to an MLP regression head. The design adapts hierarchical Transformer compression Pappagari et al. (2019); Nawrot et al. (2021) to sequence-level regression.

The reference model used chunk geometry $C = 32$, $S = 24$, embedding dimension 128, two local and two global Transformer blocks with four attention heads and feed-forward dimension 512, followed by a $128 \to 64 \to 1$ regression head.

For a 2200-nt sequence, the hierarchy reduces the global input from 2196 to 93 positions while preserving local 6-mer order and global chunk order.

4.2 Simple transformer

While the previous section presented a diagnostic-guided development of the architecture, we used also a simpler 4-layer transformer encoder model as presented in 3. The raw DNA sequence was tokenized using 6-mer tokenization (stride of 1) similarly to DNABERT (Ji et al., 2021) and prepended relevant [TISSUE] token at the beginning. The expression data were normalized and variance-corrected using min-max and Z-score normalizations based on the statistic of the training split of the data.

To predict a single continuous value out of the sequence, the model utilizes a paradigm inspired by the BERT model using the already prepended [TISSUE] token. The final predicted expression is derived entirely from regression head attached exclusively to the tissue token's output state.

To establish a foundational understanding of the Drosophila genomic grammar, the network was first pretrained on training data raw DNA sequences using Masked Language Modeling Objective and cross entropy loss. The network was pretrained through 14 epochs at 8% masking rate to help force the attention mechanism to learn local motifs and DNA

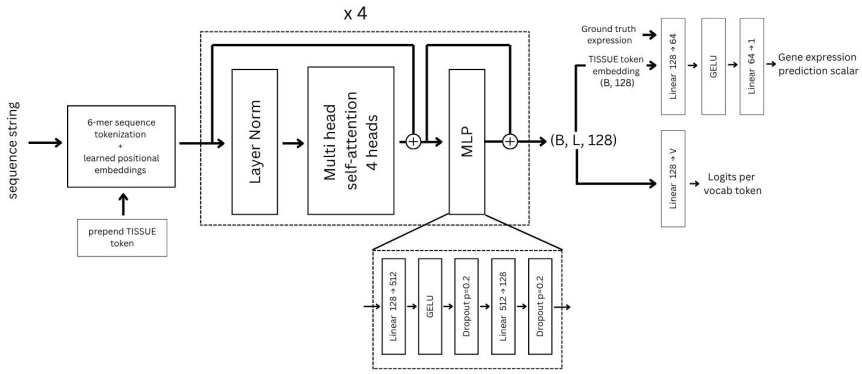


Figure 3: Architecture of the model. The L value is the batch-dependent max sequence length while $V = 4109$ is the vocabulary size. The upper head was user for training final regression model while the lower one during MLM pretraining.

syntax first.

Following the MLM pretraining, the regression head was attached to the pretrained encoder and the model was trained further to predict the expression values for a total of 35 epochs. To ensure stable gradients, we used gradient accumulation with effective batch size of 256 achieved by using micro-batch of size 8 paired with 32 gradient accumulation steps. To avoid early gradient explosion, we employed linear learning rate warmup from 1e-6 to 1e-4 over the course of 5 epochs during both MLM pretraining and training. Detailed parameters are presented in table 5 in the appendix.

5 Results and Discussion

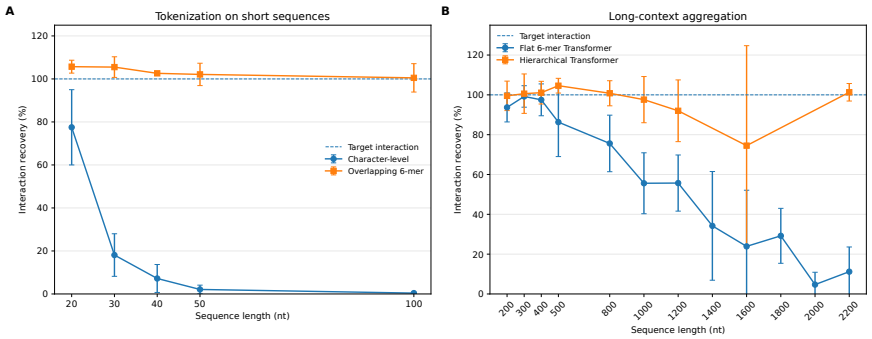


Figure 4: Interaction recovery as a function of sequence length. Panel A compares character-level and overlapping 6-mer tokenization on short inputs. Panel B compares flat and hierarchical 6-mer Transformers on longer inputs. Points show means over four seeds and error bars denote sample standard deviations. For the flat model, the best validation configuration at each length is shown.

5.1 Synthetic Scaling and Architecture Comparison

Figure 4 separates two failures: inadequate local representation in the character-level model and unreliable long-context aggregation in the flat 6-mer Transformer. Overlapping 6-mers solved the first problem, while the hierarchical model retained the synthetic interaction over 2200-nt inputs.

The flat 6-mer Transformer remained effective up to approximately 400 nt but degraded as the context increased, recovering only 11.2% of the target interaction at 2200 nt. Increasing the training budget did not systematically improve long-sequence generalization. In contrast, the hierarchical model retained high recovery over most tested lengths and reached $101.3\% \pm 4.4\%$ at 2200 nt.

The matched 2200-nt comparison in Table 1 shows that both hierarchical variants were smaller than the flat Transformer and processed substantially fewer global positions while recovering most of the known interaction. Their advantage therefore reflects local-to-global compression rather than parameter count.

Table 1: Matched comparison on synthetic 2200-nt sequences. Values are means ± sample standard deviations over four seeds.

Architecture	**Parameters**	**Global positions**	**Normalized test MSE**	**Recovery**
Flat 6-mer	1,608,193	2196	0.595 ± 0.054	$7.8\% \pm 9.0\%$
Hierarchical 16/12	1,353,217	184	$\mathbf{0.240 \pm 0.016}$	$\mathbf{95.9}\% \pm \mathbf{1.3}\%$
Hierarchical 32/24	1,343,617	93	0.311 ± 0.077	$87.7\% \pm 10.4\%$

5.2 Architectural Ablations and Input Dependence

Chunk-geometry and boundary-position ablations showed that overlap prevents motif-scale features from being lost at local boundaries; full results are reported in Appendix Figure 10.

Input interventions tested whether the hierarchical models relied on tissue identity and DNA sequence rather than on one modality alone. Figure 5 reports MSE relative to the unchanged input for each model.

Tissue corruption affected the explicitly conditioned synthetic task much more strongly than real data. All DNA perturbations increased real-data error, showing that the models use both tissue identity and gene-specific sequence information.

5.3 Results of the MLM-Pretrained Transformer

The training stability and convergence of the model are presented in Figure 6, detailing the loss curves for both the initial Masked Language Modeling (MLM) pretraining and the subsequent expression regression fine-tuning.

The model successfully minimized the Mean Squared Error (MSE), achieving a final validation MSE of 0.42. The training was deliberately halted at this point, because gene expression prediction from strictly local DNA sequence (2,200 bp) is a highly partially observable problem Avsec et al. (2021).

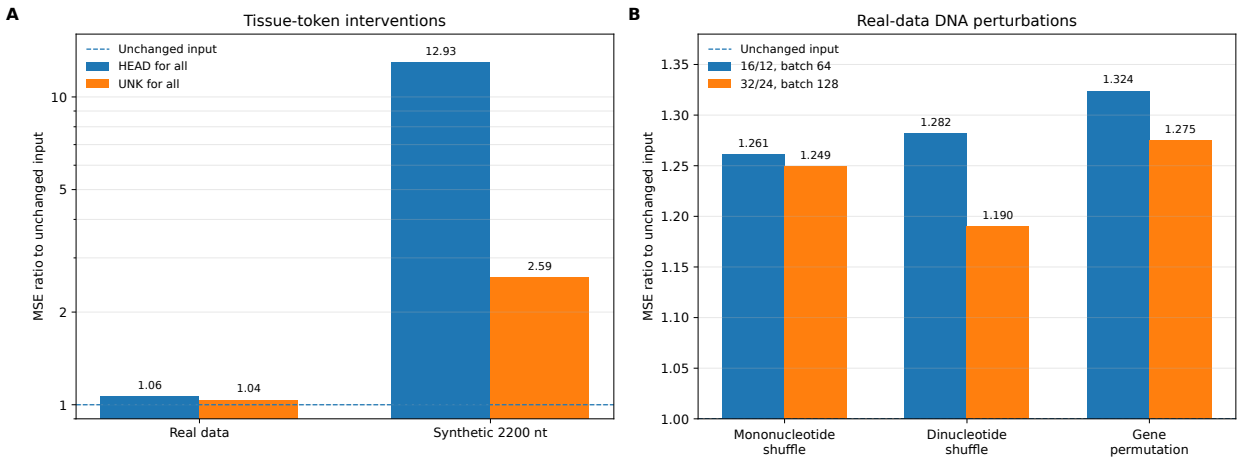


Figure 5: Dependence on tissue identity and DNA sequence. Values are normalized to the unchanged input for the same model. Panel A shows tissue-token interventions on real and synthetic data; the vertical axis is logarithmic. Panel B shows real-data sequence perturbations.

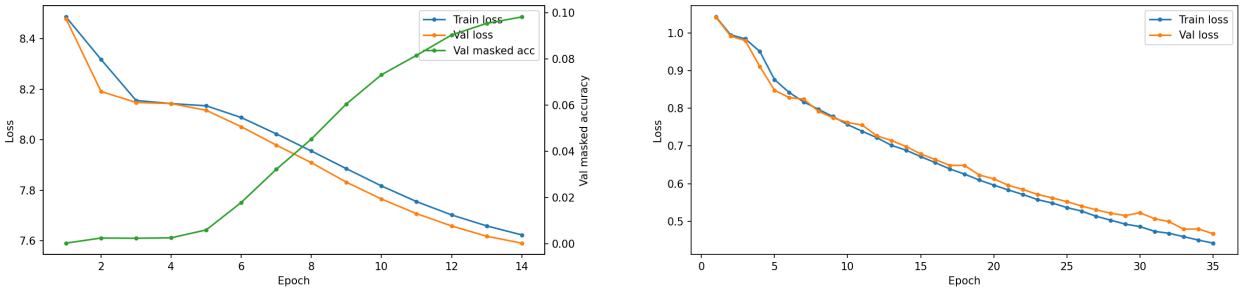


Figure 6: Training plots for (a) MLM pretraining with cross-entropy loss and (b) encoder + regression head training with MSE after MLM pretraining.

The input space (X) lacks critical multimodal biological features, such as 3D chromatin conformation, distant enhancer loops (often >50000 characters away in the genome), and epigenetic methylation states and the model, even if mathematically better, cannot achieve this without exploiting dataset noise.

Table 2: Tissue token test to check sensitivity to its value. We performed two tests on the test set except the correct tissue inference: ABDOMEN tissue to each sample and random token from the vocabulary to samples. The test used seed 42.

Experiment	14 epoch	18 epochs	35 epochs
correct	0.681037	0.638269	0.471854
the same token	0.732099	0.713119	0.559521
random string	0.732215	0.685284	0.601963
val MSE	0.69	0.62	0.42

To verify that the model actively utilized the conditional [TISSUE] token rather than relying solely on global sequence statistics, an interference test was conducted on the test set. The model's predictions were evaluated

under three conditions: a) the correct unmodified tissue token, b) a static tissue token ([TISSUE=ABDOMEN]) applied to all samples, and c) a random token sampled from the vocabulary. We observed the model succesively increases its dependence on the TISSUE token moving from marginal difference between predictions after 14 epochs to clear dependence in epoch 35 (Table 2).

After the macro test checking TISSUE token usage, we performed a micro test to establish the model's capacity to differentiate between every TISSUE pair. We evaluated predictions across all 42 ordered tissue permutations calculating, as previously, the prediction with the right TISSUE (target) and the same sequence with TISSUE modified to intervention TISSUE. On figure 8 we present the differences between those predictions across all ordered pairs. Higher values indicate tissue pairs (e.g., HEAD vs DIGESTIVE) where the model successfully established a large prediction gap, demonstrating a separation induced by the TISSUE. Lower values (e.g., ABDOMEN vs DIGESTIVE) highlight pairs for which the model assigns the same value regardless of the TISSUE which suggest it learned no clear separation between that pair.

5.4 Interpretability of the Hierarchical Transformer

5.4.1 Causal Activation Patching of Tissue Routing

Because the tissue token enters the model only at the global level, routing analysis was restricted to the two global Transformer blocks. For each ordered pair of distinct tissues, the correct tissue token was replaced with another valid token, and the resulting prediction change was measured. To localize these effects, each corrupted run was repeated while patching one global component with its clean-run activation. Ten components were tested per tissue pair: four attention heads and one MLP in each of two blocks.

Tissue-token replacement produced the largest and most consistent changes for `GONADS`-related contrasts (Figure 7A). Activation patching localized most recoverable effects to global MLPs rather than individual attention heads (Figure 7B), with median recovery of 70.7% on real data and 55.4% on synthetic data (Figure 7C). These results support a distributed, MLP-dominated routing mechanism rather than a simple one-tissue–one-head organization.

As an additional validation, attention rollout and 6-mer occlusion recovered the known synthetic motif, but produced no stable real-data motif candidate; full results are reported in Appendix 7.1.

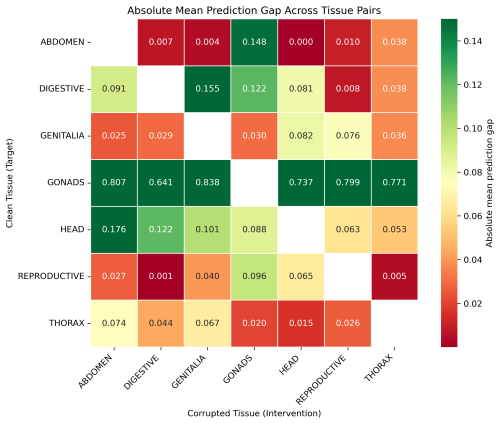


Figure 8: Model competence heatmap showing differences between prediction of sequence with target tissue and the same sequence with tissue switched to intervention tissue.

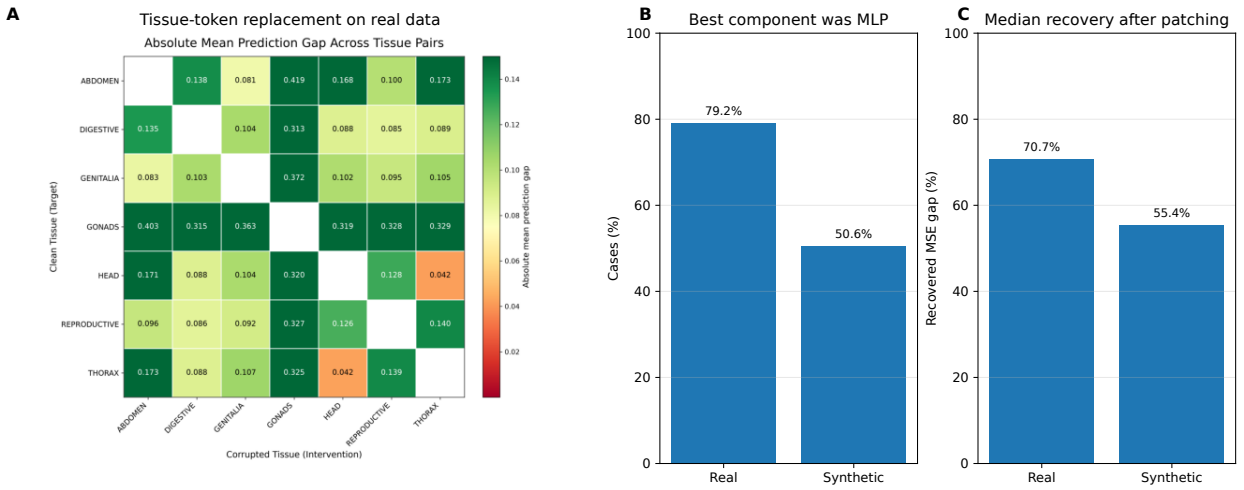


Figure 7: Summary of causal tissue-routing results. Panel A shows the mean absolute prediction change after tissue-token replacement on real data; rows denote the correct tissue and columns the intervention token. Panel B reports the percentage of pair–seed cases in which the best recovery was obtained by patching a global MLP rather than an attention head. Panel C shows the median MSE recovery over cases in which tissue-token corruption increased error.

5.5 Interpretability of the MLM-Pretrained Transformer

To identify structural elements responsible for prediction, we utilized Activation Patching. A "corrupted" forward pass is executed using an incorrect tissue token with prediction $P_{corrupt}$, followed by a "clean" pass using the correct token P_{clean}. Internal activations at a specific scan point (an attention head, or an entire block) from the clean pass are then patched into the corrupted network at this point and the network is allowed to finish the inference giving $P_{patched}$ prediction. We quantify the causal significance of a component by its Restoration Score, defined as the aggregate ratio of the repaired prediction gap:

$$Score = \frac{\sum(P_{patched} - P_{corrupt})}{\sum(P_{clean} - P_{corrupt})}.$$

We performed that detailed scan over all elements of the model for HEAD (target) vs DIGESTIVE (corruption) pair which was previously evaluated with prediction gap suggesting good separation (figure 8). The scan (Table 3, appendix) isolated Layer 4, Head 1 (L4H1) as the primary head responsible for predictions.

6 Conclusions

We succesfully trained a transformer encoder with a regression head for prediction of gene expression. The model exhibited emergence of the TISSUE token reliance and closer inspection indicates, that out of 7 tissues (Abdomen, Digestive, Genitalia, Gonads, Head, Reproductive,

Thorax) the model learned to recognize Gonads and Head. For the remaining tissues, the model has more restricted usage of the TISSUE token. This might be induced by the fact that we have a smaller model than the number of tissues (4 heads) and biologically, the most isolated head (which is a highly specialized region) and gonads (which are physically isolated from the rest of the body). The remaining tissues have more blended borders which suggests why the model focused on those specific ones.

The hierarchical experiments showed that low prediction error alone did not guarantee recovery of the intended tissue x motif interaction. Overlapping 6-mers improved local motif recognition, while hierarchical local-to-global compression preserved the interaction on 2200 nt sequences and outperformed the flat Transformer despite using fewer parameters and substantially fewer global positions. The boundary-position experiment further showed that overlapping chunks protect motif-scale information near chunk's boundaries.

On real data, tissue-token and DNA perturbations confirmed that the hierarchical models used both tissue identity and gene-specific sequence information. Activation patching suggested distributed, MLP-dominated routing rather than specialization in a single attention head. Rollout and occlusion recovered the known synthetic motif, but no stable real-data motif, so the real-data results should be interpreted as properties of the trained model rather than evidence of causal biological mechanisms.

7 Appendix

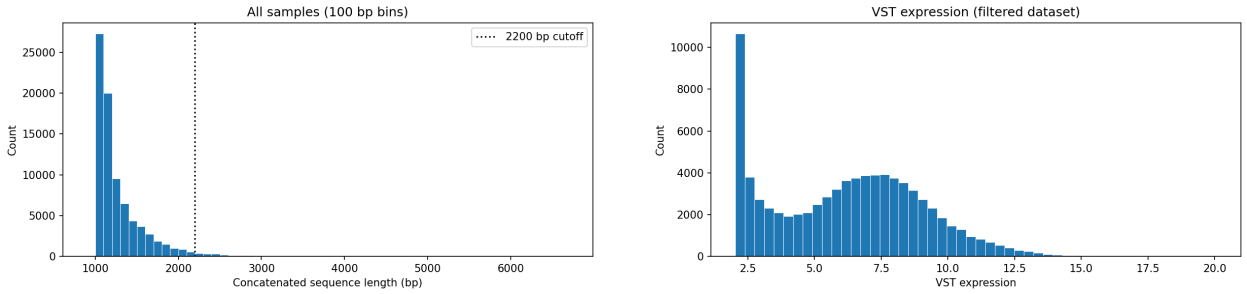


Figure 9: Dataset stats (a) sequence string length histogram (bin 100 characters) and (b) expression counts distribution. The VST expression counts are skewed towards small values which is expected for a gene expression dataset.

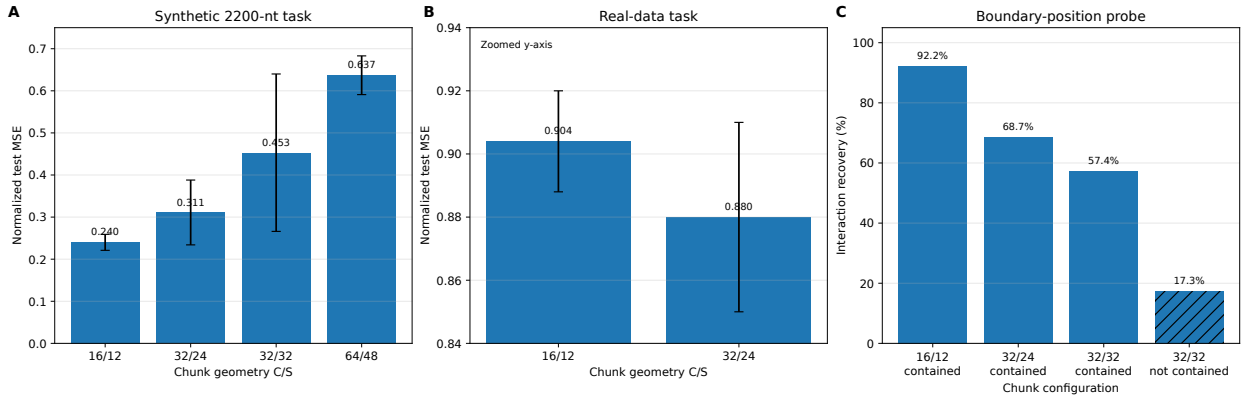


Figure 10: Effects of chunk geometry and overlap. Panels A and B compare held-out normalized MSE across synthetic and real data. Panel C shows the controlled boundary-position probe. Interaction recovery decreases sharply when a motif crosses a boundary in the non-overlapping model and is not fully contained in either chunk.

7.1 Hierarchical Attention Rollout and K-mer Occlusion

Because the tissue token interacts with DNA only through chunk summaries, token-level saliency was obtained by combining global tissue-to-chunk rollout with local `[CHUNK]`-to-6-mer rollout. For position j in chunk i,

$$S_{i,j} = S_i^{\text{global}} S_{i,j}^{\text{local}}.$$

Scores from overlapping chunks were projected back to 6-mer positions. To test whether highly ranked tokens were causally important, the ten highest-scoring unique 6-mers per seed were replaced with the unknown token and compared with random controls.

Table 3: Restoration scores for case HEAD v DIGESTIVE across all model componends. We see high score in L4H1.

HEAD (target) vs DIGESTIVE (intervention)							
block	component	head	score	block	component	head	score
1	attention	0	0.068533	3	attention	0	-0.01225
1	attention	1	-0.190864	3	attention	1	-0.112725
1	attention	2	0.057667	3	attention	2	-0.028954
1	attention	3	0.050479	3	attention	3	-0.007395
2	attention	0	0.000466	4	attention	0	0.094068
2	attention	1	-0.060283	4	attention	1	0.53805
2	attention	2	-0.009176	4	attention	2	0.095729
2	attention	3	-0.020294	4	attention	3	0.110201
1	mlp	<NA>	0.086173	3	mlp	<NA>	-0.097975
2	mlp	<NA>	0.243195	4	mlp	<NA>	0.195437

Synthetic positive control. On HEAD samples containing `GATTACAA`, rollout recovered all three motif-derived 6-mers in every seed. Table 4 summarizes the corresponding occlusion effects.

Table 4: Recovery of the known synthetic motif using hierarchical attention rollout and causal 6-mer occlusion. MSE increase is measured relative to the unmodified input.

6-mer	Rollout seeds	Mean rollout score	Occlusion seeds	Mean normalized MSE increase
`TTACAA`	4	1.570	4	7.676
`GATTAC`	4	0.695	3	8.124
`ATTACA`	4	0.345	4	0.645

Occluding `GATTAC` or `TTACAA` strongly increased MSE, whereas `ATTACA` had a smaller and geometry-dependent effect. The agreement between rollout and occlusion validates the method on a task with known ground truth and shows that the model relies on motif-derived sequence information, although not uniformly across overlapping 6-mers.

Real-data results. For real HEAD samples, rollout mainly highlighted low-complexity 6-mers such as `AAAAAA` and `TTTTTT`. Their occlusion effects were small, inconsistent across seeds, and comparable to random controls. Thus, unlike the synthetic positive control, the real-data analysis did not identify a stable candidate motif and does not support a specific biological interpretation.

Table 5: Parameters used for training of the simple transformer with MLM-pretraining. For all training and downstream tasks we used seed=42.

Data			
vocab_pad	[PAD]	kmer_size	6
vocab_mask	[MASK]	kmer_stride	1
vocab_unk	N	max_total_seq_len	2200
nucleotides	['A', 'C', 'T', 'G']		
Training general			
batch_size	8	warmup_period	5
gradient_accumulation_steps	32	val_split	0.1
train_split	0.8	test_split	0.1
Pretraining			
mask_percent	0.08	mlm_max_lr	1.00E-04
mlm_epochs	11	mlm_warmup_period	5
mlm_start_lr	1.00E-06	mlm_validation_split	0.1
Training			
finetune_start_lr	1.00E-07	epochs	35
finetune_max_lr	5.00E-05		

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